Bayesian forecasting of the return volatility with Markov Switching GARCH (MS-GARCH) and Stochastic Volatility (SV)

Introduction

- The variance forecasting is central whenever an optimal decision is a function of the predictive distribution that cannot be summarized by a point estimate.
- An example is the mean-variance tradeoff in finance and insurance.
- Most variance models are not structural and may be selected based on the out-of-sample performance.
- MS-GARCH and SV can create most of the stylized facts observed in the return data and may look marginally similar (Figure 1).

I compare the forecasting performance of two conditional variance models, MS-GARCH and SV, on the S&P 500 daily returns for 2016-2026.

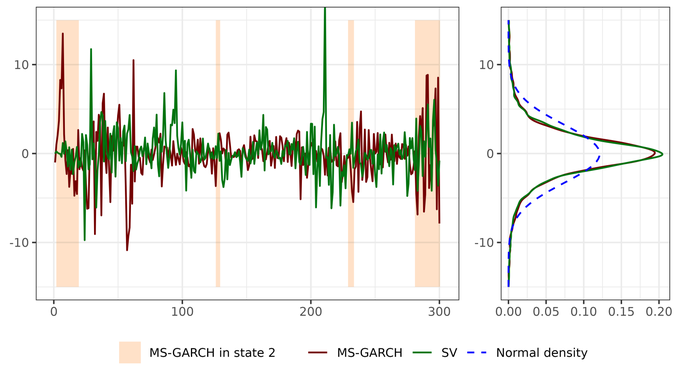


Figure 1: Example. Simulated MS-GARCH and SV processes with matched parameters.

MS-GARCH model

$$
\begin{aligned}
y_t &= \sigma_t z_t \quad z_t \sim \mathcal{N}(0,1) \\
\sigma_t^2 &= \omega_{s_t} + \alpha_{s_t} y_{t-1}^2 + \beta_{s_t} \sigma_{t-1}^2 \\
w_k \geq 0, \alpha_k \geq 0, \beta_k &\geq 0 \quad \text{for regimes } k \in \{1,2\}
\end{aligned}
$$

where s_t is a first order Markov chain process with the domain $\{1,2\}$

SV model

$$
\begin{aligned}
y_t &= \sqrt{h_t} z_t, \quad z_t \sim \mathcal{N}(0,1) \\
\log h_t &= \mu + \varphi \log h_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \sigma_u^2) \quad z_t \perp u_t
\end{aligned}
$$

Estimation and forecasting

Both models assume non-linear dynamics that is hard to translate into a likelihood. Standard MS-GARCH is considered intractable because evaluation of the likelihood requires summing over all possible state paths 2^T. SV is intractable because it requires T dimensional integration of the whole history of the latent volatility. Filtration is not feasible in both cases.

To deal with it, the models are altered in subtle ways. The R package I use to estimate MS-GARCH (Ardia et al., 2019) assumes parallel within-regime volatility processes:

$$
\begin{aligned}
y_t &= \sigma_{(s_t,t)} z_t \quad z_t \sim \mathcal{N}(0,1) \\
\sigma_{(1,t)}^2 &= \omega_1 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{(1,t-1)}^2 \\
\sigma_{(2,t)}^2 &= \omega_2 + \alpha_2 y_{t-1}^2 + \beta_2 \sigma_{(2,t-1)}^2.
\end{aligned}
$$

The R package I use to estimate SV (Kastner, 2016) uses the normal mixture approximation:

$$
\begin{aligned}
y_t &= \sqrt{h_t} z_t \\
\Leftrightarrow \log y_t^2 &= \log h_t + \log z_t^2, \quad z_t \sim \mathcal{N}(0,1) \\
&\approx \log h_t + e_t, \quad e_t \mid r_t \sim \mathcal{N}\left(\mu_{r_t}, \gamma_{r_t}^2\right), \quad r_t \in \{1, ..., 10\}
\end{aligned}
$$

where r_t is a mixture component indicator at time t. For both models the estimation is Bayesian MCMC with the default uninformative priors.

For the forecast I use the expanding window estimation and reserve the testing window $y_{S:T}$ consisting of 500 observations. Specifically, a model is refitted to the $y_{1:t}$ at every time point $t \in \{S-1, ..., T-1\}$. Once the posterior sample of parameters $\left\{\theta^{(m)}\right\}_{m=1}^{M}$ is obtained for a given fit, forecasting is done by simulating the process with each parameter draw. The result is the collection of draws from the predictive distribution

$$
\left\{\sigma_{t+d}^{(m)}, y_{t+d}^{(m)}\right\}_{m=1}^{M} \text{ for every desired horizon } d.
$$

Forecast evaluation

Let A, B denote MS-GARCH and SV, d the number of days ahead in the forecast, and p the predictive density defined by a model passed in brackets. Then, the Log-Likelihood score function is

$$
\mathrm{L}(A, B, d) = \sum_{t=S}^{T} \log\left[\frac{p(y_{t+d} \mid y_{1:t-1}, A)}{p(y_{t+d} \mid y_{1:t-1}, B)}\right].
$$

The predictive distribution for a model X can be evaluated by one of the three methods:

$$
p(y_{t+d} \mid y_{1:t-1}, X) = \begin{cases}
\frac{1}{M} \sum_{m=1}^{M} \varphi\left(y_{t+d}, 0, \sigma_{t+d}^{2(m)}\right) & \text{(Model-Implied density)} \\
\frac{1}{M b_{t-1}} \sum_{m=1}^{M} \varphi\left(\frac{y_{t+d} - y_{t+d}^{(m)}}{b_{t-1}}\right) & \text{(Gaussian KDE)} \\
\frac{1}{M b_{t-1}} \sum_{m=1}^{M} t_{\nu=2}\left(\frac{y_{t+d} - y_{t+d}^{(m)}}{b_{t-1}}\right) & \text{(Student-t KDE)}
\end{cases}
$$

where b_{t-1} is the SJ bandwidth computed using $y_{1:t-1}$.

Another score function concerns a volatility point forecast, requiring the realized volatility measure (RV) as a proxy for the variance:

$$
\mathrm{QLIKE}(X, d) = -\sum_{t=S}^{T} \left(\log(\hat{\sigma}_{t+d}^2) + \frac{\mathrm{RV}_{t+d}}{\hat{\sigma}_{t+d}^2}\right).
$$

The intraday observations are used to compute RV_{t+d}. I define a point forecasts to be the mean of the predictive: $\hat{\sigma}_{t+d} = \overline{\sigma_{t+d}^{(m)}}$.

Similarly to the log-likelihoods, two models are compared with the quantity

$$
\mathrm{QLIKE}(A, B, d) = \mathrm{QLIKE}(A, d) - \mathrm{QLIKE}(B, d).
$$

The QLIKE is supposed to be more robust because it uses additional information to estimate the volatility ex-post.

Results 😲

- SV scores higher than MS-GARCH on longer horizons, while for shorter horizons scores disagree. For example, see QLIKE in Figure 2.
- The scores are consistent for their targets but are heavily affected by the outliers because of the (wrong) conditional Gaussian assumption.
- As a consequence, different testing window would lead to different conclusions.

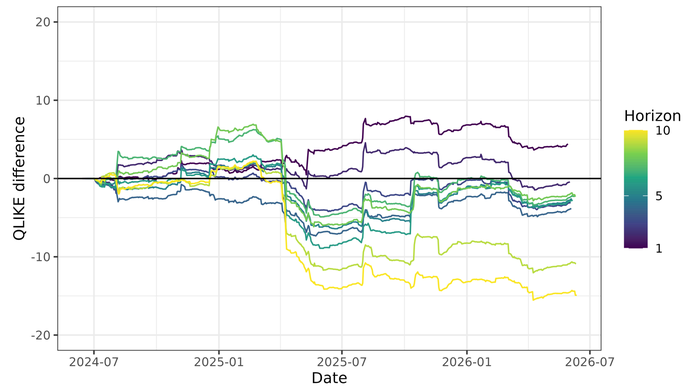


Figure 2: The difference in the cumulative QLIKE. Positive values favor MS-GARCH.

A bigger issue with MS-GARCH

- Figure 3 shows the draws from the 5 days ahead volatility predictive distribution, where MS-GARCH features two dense regions corresponding to two regimes.
- However these is no visible regime persistence pertinent to a Markov switching.
- Furthermore (not shown here), the shapes of the parameter posteriors of the second regime depend on the seed, and MCMC traceplots indicate the divergence of chains.

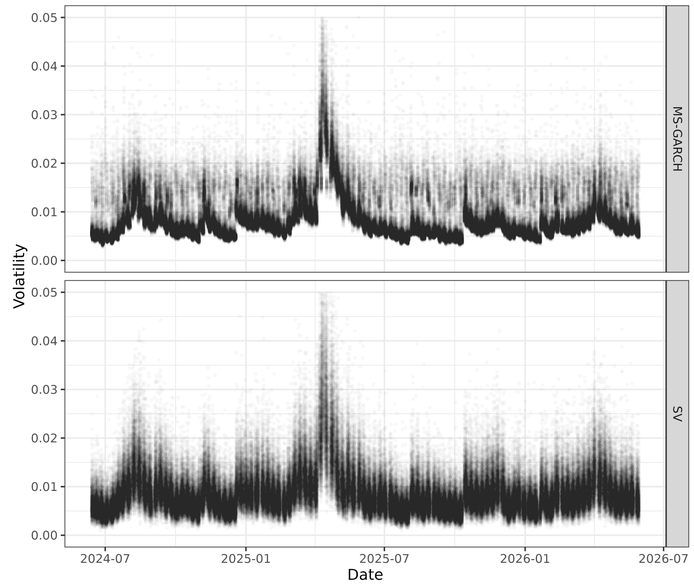


Figure 3: A scatter plot of the draws from the 5 days ahead predictive distributions $\left\{\sigma_t^{(m)}\right\}_{m=1}^{1000}$.

Conclusion ✍️

- SV is a safer default choice for S&P500 volatility modeling than MS-GARCH.
- MS-GARCH struggles to identify the regimes, resulting in the estimation instability.
- Conditional normal assumption is not consistent with the observations, resulting in scoring rules that are sensitive to outliers within a testing window.

R-packages used for estimation and forecasting:

Ardia, D., Bluteau, K., Boudt, K., Catania, L., & Trottier, D.-A. (2019). Markov-switching GARCH models in R: The MSGARCH package. *Journal of Statistical Software*, *91*, 1–38. https://doi.org/http://doi.org/10.18637/jss.v069.i05

Kastner, G. (2016). Dealing with Stochastic Volatility in Time Series Using the R Package stochvol. *Journal of Statistical Software*, *69*(5), 1–30. https://doi.org/10.18637/jss.v069.i05